

Density Ratio Estimation as Supervised Learning

Roshni Sahoo
r.sahoo@columbia.edu

Columbia University

1 Introduction

Density ratio estimation is a fundamental problem in causal inference and learning under distribution shift. The most common approach to density ratio estimation is probabilistic classification [Menon and Ong, 2016, Sugiyama et al., 2008]. In probabilistic classification, we observe i.i.d. samples $X \sim P$ and $X \sim Q$. We provide pseudolabel $Y = 1$ to samples drawn from Q and pseudolabel $Y = 0$ to samples drawn from P . Let F denote the joint mixture distribution over (X, Y) with these labels. Then, we fit a classifier $h(x) = \mathbb{P}[Y = 1 \mid X]$. By Bayes' rule, we can show that the density ratio $r(x) := \frac{dQ_X(x)}{dP_X(x)}$ is given by

$$r(x) = \frac{h(x)}{1 - h(x)}.$$

The problem with this approach to density ratio estimation is that it can become unstable when classification probabilities approach 1. Suppose that the classifier is estimated with some error $\hat{h}(x)$. Let $\delta(x) := \hat{h}(x) - h(x)$. We can compute the error in $r(x)$ that arises from the classifier error:

$$\begin{aligned} |\hat{r}(x) - r(x)| &= \left| \frac{\hat{h}(x)}{1 - \hat{h}(x)} - \frac{h(x)}{1 - h(x)} \right| \\ &= \left| \frac{\delta(x) \cdot (1 + r(x))^2}{1 - \delta(x) \cdot (1 + r(x))} \right|. \end{aligned}$$

Rather than estimate the classifier and then form the density ratio, we propose to directly estimate the density ratio by solving a loss minimization problem

$$\min_{r: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_F [L_\psi(r(X), Y)]. \quad (1)$$

Define $\psi : (0, \infty) \rightarrow \mathbb{R}$. Define $\phi(r) = \int_0^r t\psi'(t)dt$. More broadly, we can consider the family of loss functions given below.

$$L_\psi(r, y) = \phi(r) \cdot (1 - y) - \psi(r) \cdot y. \quad (2)$$

Assumption 1. The function ψ is twice-continuously differentiable, strictly increasing, concave, and has the property that $\psi'(r) + r\psi''(r) \geq 0$.

Lemma 1. Let $X_i \sim P$ denote i.i.d. samples from P and $X_j \sim Q$ denote i.i.d. samples from Q . Pseudolabel the samples from P by $Y_i = 0$ and pseudolabel the samples from Q by $Y_i = 1$. The loss function L_ψ given in (2) is convex in r . In addition, (1) is strictly convex and has a unique minimizer at the true density ratio $r(x) = \frac{dQ_X(x)}{dP_X(x)}$.

Proof. We note that

$$\begin{aligned} L'_\psi(r, y) &= \phi'(r) \cdot (1 - y) - \psi'(r) \cdot y \\ &= r\psi'(r) \cdot (1 - y) - \psi'(r) \cdot y. \end{aligned}$$

In addition, we have that

$$\begin{aligned} L''_\psi(r, y) &= \phi''(r) \cdot (1 - y) - \psi''(r) \cdot y \\ &= (r\psi''(r) + \psi'(r)) \cdot (1 - y) - \psi''(r) \cdot y \end{aligned}$$

Since ψ is concave, we have that $L''_\psi(r, 1) \geq 0$. Since $r\psi''(r) + \psi'(r) \geq 0$, we have that $L''_\psi(r, 0) \geq 0$. Thus, L_ψ is convex in r .

We show that the unique population minimizer occurs at $r(x) = \frac{dQ_X(x)}{dP_X(x)}$. We have that the first-order condition of the population risk is given by

$$\begin{aligned} \nabla_{r(x)} \mathbb{E}[L_\psi(r(X), Y) \mid X = x] &= r(x) \cdot \psi'(r(x)) \cdot \mathbb{P}[Y = 0 \mid X = x] - \psi'(r(x)) \cdot \mathbb{P}[Y = 1 \mid X = x] \\ &= \psi'(r(x)) \cdot (r(x) \cdot \mathbb{P}[Y = 0 \mid X = x] - \mathbb{P}[Y = 1 \mid X = x]). \end{aligned}$$

Since ψ is strictly increasing, $\psi'(r) > 0$, so the unique stationary point of the population risk is

$$r(x) = \frac{\mathbb{P}[Y = 1 \mid X = x]}{\mathbb{P}[Y = 0 \mid X = x]} = \frac{\mathbb{P}[X = x \mid Y = 1]}{\mathbb{P}[X = x \mid Y = 0]} = \frac{dQ_X(x)}{dP_X(x)},$$

where the last equalities follow by Bayes' rule. □

References

- Aditya Menon and Cheng Soon Ong. Linking losses for density ratio and class-probability estimation. In *International Conference on Machine Learning*, pages 304–313. PMLR, 2016.
- Masashi Sugiyama, Taiji Suzuki, Shinichi Nakajima, Hisashi Kashima, Paul Von Büna, and Motoaki Kawanabe. Direct importance estimation for covariate shift adaptation. *Annals of the Institute of Statistical Mathematics*, 60:699–746, 2008.